

Do Discounts Influence Consumer Spending Behavior?

In this project, I aim to analyze how consumer spending behavior changes during large promotional events such as Black Friday. I will use a publicly available retail dataset that includes customer information such as age, gender, occupation, and city category, along with product categories and purchase amounts. The dataset contains a large number of observations, which makes it suitable for both statistical analysis and basic machine learning methods.

Since the dataset does not directly include discount rates, I will use proxy variables such as product categories, total spending, and purchasing patterns to understand the possible effects of discounts. I will also group and analyze customers based on demographic variables to see how different types of consumers behave in these situations. In addition, to enrich the dataset, I will create new features such as price levels derived from product categories and regrouped demographic variables (for example, broader age groups). These derived features will help capture underlying patterns in consumer behavior and strengthen the analysis.

The project will include data cleaning, exploratory data analysis (EDA), and simple hypothesis testing to examine relationships between variables such as age, gender, and spending. In addition, I plan to apply a regression model to predict purchase amounts based on these features.

This topic is interesting to me because I have previously done a presentation on a similar subject in one of my previous courses. During that presentation, I became interested in how marketing strategies, especially discounts, can influence people's purchasing decisions. Therefore, I would like to explore this topic further using real data and apply data science techniques to better understand consumer behavior.